

Module 3: Shell Hierarchy and the Nested Fold Field

From Microscopic Shell Formation to Cosmological Structure

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Abstract

Module 1 established that three-dimensional space emerges when regions of the Fold Field close upon themselves, forming shells. This paper extends that result upward in scale: the same shell-formation dynamics that produced 3D geometry operate at all scales, organizing matter into a nested hierarchy of gravitationally bound shells. We derive the fold-native shell boundary condition — the Lagrange surface where one fold field yields to another — and show how the background structural mass term B transforms across shell boundaries. The universe itself is the outermost shell, closed during the 2D→3D transition. The differential $B_{\text{boundary}} - B_{\text{interior}}$, built up as structure forms within the cosmological shell, is identified as a candidate mechanism for dark energy. The timing is natural: the dark sector cannot appear before shell closure, which is why it arrives late. All derivations use fold primitives $(\Phi, \bar{\tau}, \kappa)$ and the results of Papers 1–4 in the PFT series.

1 Introduction

The dimensional ladder of Module 1 runs upward from the Prime Substrate:

$$0\text{D} \rightarrow 1\text{D} \rightarrow 2\text{D (Fold Field)} \rightarrow 3\text{D (Shells)}. \quad (1)$$

Shells are the mechanism by which dimensionality crystallizes: a region of the Fold Field closes upon itself, acquiring interior/exterior structure and quasi-radial layering. Once shells exist, gravity follows (Paper 3 [7]) and gauge structure organizes (Papers 1–2 [5, 6]).

That account stops at the Planck scale. It explains how 3D space emerged, but not how matter organized itself into stars, planets, and galaxies within that space. This paper argues the gap is not a new mechanism but the same one: shell formation continues operating at every scale above the Planck length, producing a nested hierarchy that extends from subatomic to cosmological scales.

1.1 The Central Claim

We hypothesize that the fold field is scale-invariant in its shell-formation dynamics: a region of the fold field closes into a shell whenever its local $g_{\text{local}}(r) > g_{\text{external}}(r)$, regardless of scale, and the shell boundary is the surface where these fields cross. The boundary condition is derived; scale invariance of the full dynamics is a hypothesis supported by the absence of explicit scale dependence, not yet a proven theorem.

The boundary condition itself follows from the inverse-square law (Paper 3) and the superposition principle (which holds exactly in the static limit) — these are results, not postulates. The claim that this same condition applies at all scales is the hypothesis. The novelty is in tracing its consequences: a nested B -hierarchy, screening of the cosmological background, and a natural candidate for dark energy.

1.2 What This Paper Does Not Claim

- We do not derive the magnitude of dark energy from fold primitives. That requires a fold cosmology not yet developed.
- We do not prove scale invariance of shell formation. We derive the boundary condition and note that it has no explicit scale dependence; we flag this as a hypothesis requiring further verification.
- We do not identify the dark matter candidate precisely. We note the natural slot for a high- κ fold sector and defer its development.

2 Shell Formation: From Planck Scale to Macroscopic

2.1 The Module 1 Shell

Module 1 defines a shell as a closed region of the Fold Field with:

1. A defined interior/exterior boundary
2. Quasi-radial layering (the fold density Φ stratifies by distance from the shell center)
3. Net skew $\mathcal{S} = \nabla \cdot g \neq 0$ at the center (a mass source in the sense of Paper 3)

Three-dimensional space itself is such a shell — the outermost one, formed during the 2D→3D transition driven by cost-of-connectivity dynamics (spectral dimension $d_s \approx 3.1$, confirmed numerically).

2.2 The Macroscopic Shell

At scales above the Planck length, the fold field has already crystallized into 3D geometry. A macroscopic object (planet, star, galaxy) is a region where the local fold density Φ and structural mass $m = \Phi\bar{\tau}$ are elevated above the background. The gravitational field of this object:

$$g_{\text{local}}(r) = \frac{A_{\text{local}}}{r^2} \quad (2)$$

competes with the gravitational field of any containing object:

$$g_{\text{ext}}(r) = \frac{A_{\text{ext}}}{(d-r)^2} \quad (3)$$

where d is the distance to the external source.

Definition 1 (Shell Boundary). *The shell boundary of a fold object with structural mass parameter A_{local} within an external field with parameter A_{ext} at distance d is the surface $r = r^*$ satisfying:*

$$\frac{A_{\text{local}}}{r^{*2}} = \frac{A_{\text{ext}}}{(d-r^*)^2}. \quad (4)$$

Inside r^ : local fold dynamics dominate. Outside r^* : external fold dynamics dominate.*

This is the fold-native form of the Hill sphere / Lagrange surface. It is not a postulate. Module 2.3 [4] (Section 4) derives it analytically from $1/r^2$ and superposition, and confirms it numerically on the fold lattice with correlation $r = 0.994$ across five mass ratios (errors $< 6\%$, consistent with lattice discretization). The closed-form solution is:

$$r^* = \frac{d}{1 + \sqrt{A_{\text{ext}}/A_{\text{local}}}}, \quad (5)$$

which follows from setting $g_{\text{local}}(r^*) = g_{\text{ext}}(r^*)$ and solving. The larger source dominates the larger volume — “bias to the larger” is a mathematical consequence of the fold field, not an additional assumption.

Proposition 1 (Scale Independence of Boundary Condition). *Equation (4) contains no explicit length scale. The shell boundary exists at any scale where a local fold field competes with an external one, from nuclear to cosmological.*

Nondimensionalization argument. To verify that scale invariance is not merely absence of explicit scale but a structural property, we nondimensionalize the fold field equations. Let r_0 be any reference length, m_0 any reference mass, $t_0 = r_0/c$ the associated time. Define dimensionless variables:

$$\tilde{r} = r/r_0, \quad \tilde{m} = m/m_0, \quad \tilde{t} = t/t_0, \quad \tilde{\tau} = \bar{\tau}/t_0, \quad \tilde{\kappa} = \kappa/t_0. \quad (6)$$

The fold transport equation, accumulation equation, and shell boundary condition (4) all transform to equations in tilded variables with *identical form* and no residual dependence on r_0 , m_0 , or t_0 . The ratio $\tilde{A}/\tilde{A}_{\text{ext}}$ determining r^*/d is invariant under this rescaling.

This confirms that the shell boundary condition is not merely lacking an explicit scale — it is genuinely scale-free. The same condition applies at Planck scale, planetary scale, and cosmological scale because no combination of r_0 , m_0 , t_0 appears in the rescaled equations.

Caveat: scale invariance of the boundary condition does not guarantee scale invariance of the full dynamical solution. Boundary conditions and initial conditions may break the symmetry at specific scales. This is Open Problem 2 (Section ??).

2.3 The Rocket Thought Experiment

A rocket launching from Earth illustrates the shell hierarchy concretely. Inside Earth’s Hill sphere ($r < r_{\text{Earth}}^*$): the rocket’s fold field contributes to Earth’s $M_{\text{structure}}$ and is screened by Earth’s shell. The rocket has no independent shell.

As the rocket crosses r_{Earth}^* : the local/external field balance reverses. The rocket’s fold field now dominates locally. A new shell boundary forms around the rocket, with the rocket’s A_{rocket} competing against Earth’s field at the new distance. The rocket has claimed its own shell.

The boundary is not determined by the rocket’s decision or velocity. It is determined by the ratio $A_{\text{local}}/A_{\text{ext}}$ — bias toward the larger. A more massive object always wins within its Hill sphere.

3 The B -Hierarchy: Screening and the Nested Background

3.1 B in the General Solution

The static fold field equation $\nabla^2 m = 0$ has general solution:

$$m(r) = \frac{A}{r} + B. \quad (7)$$

B is spatially uniform within its domain, drops out of the local force law ($\nabla B = 0$), and is invisible to local dynamics. In Paper 3 this was noted as a cosmological constant analog. The shell hierarchy gives it a precise meaning.

3.2 B Transforms Across Shell Boundaries

Inside a shell at scale ℓ , the relevant background is not the cosmological B_{universe} but the local B_ℓ set by the shell’s own T3 conservation:

$$B_\ell = \frac{M_{\text{total},\ell} - M_{\text{structure},\ell}}{V_\ell}. \quad (8)$$

Derivation of screening from Definition 2.1. By Definition 2.1, inside the shell boundary $r < r^*$, the local fold field $g_{\text{local}}(r)$ dominates over the external field $g_{\text{ext}}(r)$.

The general solution $m(r) = A/r + B$ applies separately in each domain: inside the shell, $m_{\text{in}}(r) = A_{\text{local}}/r + B_{\ell}$; outside, $m_{\text{out}}(r) = A_{\text{ext}}/r + B_{\text{universe}}$. Continuity of m at r^* requires:

$$\frac{A_{\text{local}}}{r^*} + B_{\ell} = \frac{A_{\text{ext}}}{r^*} + B_{\text{universe}}. \quad (9)$$

Since $g_{\text{local}}(r^*) = g_{\text{ext}}(r^*)$ at the boundary (by Definition 2.1), the A/r terms match and we require B_{ℓ} to absorb any residual — it is set by T3 within the shell volume, not inherited from the exterior. The cosmological B_{universe} does not penetrate: it is present at the boundary but is replaced by B_{ℓ} in the interior domain, because T3 conservation within V_{ℓ} independently fixes the local mass budget. Screening is therefore a consequence of the boundary condition (Definition 2.1) combined with T3 — not of T3 alone.

Proposition 2 (*B*-Screening). *Within a fold shell of volume V_{ℓ} , the effective background structural mass is B_{ℓ} given by Eq. (8), derived from the shell boundary condition (Definition 2.1) and T3 conservation. The cosmological background B_{universe} does not contribute to local dynamics inside the shell.*

This explains why the cosmological constant does not affect planetary orbits, solar system dynamics, or galactic rotation curves at small radii — the local shell screens it completely. Dark energy only acts on objects that have exited all containing shells, i.e. objects in pure Hubble flow.

3.3 The Nested Hierarchy

The full *B*-hierarchy is:

$$B_{\text{universe}} \supset B_{\text{cluster}} \supset B_{\text{galaxy}} \supset B_{\text{solar}} \supset B_{\text{planet}} \supset \dots \quad (10)$$

Each level screens the one above. At each boundary crossing, Eq. (8) sets the new local B from the enclosed mass budget.

Dynamical hierarchy formation. A new shell boundary forms at scale L when the local fold accumulation $\bar{\tau}$ reaches threshold κ across a region of that scale, triggering a collapse event that concentrates structural mass and establishes an interior/exterior contrast. The interior B then decouples from the external field via T3 conservation (Eq. 8), and a new screening layer is established.

In plain language: *a new shell forms when local tension reaches its breaking point at a given scale, concentrating mass, drawing a new boundary, and draining the background field inside it.* The scale at which this occurs is set by the ratio κ/α and the local gradient $|\nabla m|$ — the same parameters that set the shell plateau $C = 1.014$ in Module 2.3. Each level of the hierarchy is the fold dynamics solving the same fixed-point problem at a different scale.

4 The Cosmological Shell and Dark Energy

4.1 The Universe as the Outermost Shell

The universe is the outermost fold shell — the one that closed during the 2D→3D transition. Its formation is not an event in time within the universe; it is the event that produced time

(via the Planck tick, Paper 4 [8]) and 3D space (via cost-of-connectivity dynamics). There is no containing shell above it.

Consequently, the universe's B_{boundary} has no external reference to screen it. It is set by the initial conditions at shell closure — the total structural mass of the universe at the moment of 3D crystallization.

4.2 The B -Differential and Dark Energy

As structure forms within the cosmological shell, $M_{\text{structure}}$ grows. By Eq. (8):

$$B_{\text{interior}}(t) = \frac{M_{\text{total}} - M_{\text{structure}}(t)}{V(t)}. \quad (11)$$

B_{interior} decreases as structure grows. The shell boundary B_{boundary} , set at the 2D→3D transition, does not decrease at the same rate — it is anchored to the closed-shell conditions.

The differential:

$$\Delta B(t) = B_{\text{boundary}} - B_{\text{interior}}(t) \geq 0 \quad (12)$$

grows monotonically as structure forms. In the HVAC analogy: the interior is at lower structural-mass pressure than the boundary. The boundary does not need to be negative in absolute terms — only higher than the interior. This is suction, not absolute negative pressure.

Proposition 3 (ΔB as Effective Negative Pressure). *In a closed fold system ($T3$ conservation, shell boundary fixed), the differential $\Delta B = B_{\text{boundary}} - B_{\text{interior}} > 0$ acts as an effective outward pressure on the interior. It does not require $B < 0$ and does not violate $T1$. It requires only that B_{boundary} decreases more slowly than B_{interior} as structure forms.*

4.3 Why the Dark Sector Arrives Late

The cosmological ΔB cannot exist before shell closure. Before the 2D→3D transition, there is no shell boundary, no interior/exterior distinction, and no B -differential. The dark sector — whatever manifests ΔB as observable dark energy — is structurally forbidden from appearing before 3D space exists.

After shell closure, $\Delta B = 0$ initially (no structure has formed). ΔB grows as $M_{\text{structure}}$ grows. Dark energy becomes dynamically significant when ΔB becomes comparable to the local gravitational energy density — which happens around the epoch of matter-energy equality, consistent with observations ($z \approx 0.4$).

This resolves the coincidence problem naturally: dark energy and matter are comparable today because dark energy is *sourced by* matter clustering. They are not independent quantities that happen to coincide.

4.4 Equation of State

The equation of state of ΔB depends on how B_{boundary} and B_{interior} evolve with the expansion scale factor a .

If $B_{\text{boundary}} = \text{const}$ (boundary frozen at shell-closure conditions) and $B_{\text{interior}} \propto a^{-3}$ (dilutes with volume):

$$\Delta B(a) \approx B_{\text{boundary}} - B_0 a^{-3} \xrightarrow{a \rightarrow \infty} B_{\text{boundary}} = \text{const}. \quad (13)$$

At late times $\Delta B \rightarrow \text{const}$, giving $w = -1$. At early times $\Delta B \approx B_{\text{boundary}} - B_0 a^{-3}$ is smaller and growing, giving $w > -1$ (quintessence-like).

The transition from $w > -1$ (early) to $w \rightarrow -1$ (late) is a prediction: dark energy should appear to strengthen slightly over cosmic time as B_{interior} depletes. Current observations (DESI 2024 BAO results) are consistent with w slightly above -1 and possibly time-varying, which is consistent with this picture.

4.5 What This Is Not

This is a structural identification, not a derivation of the dark energy density. The magnitude of B_{boundary} — why it corresponds to $\sim 10^{-29} \text{ g/cm}^3$ — requires a fold cosmology that connects the Planck-scale shell closure conditions to the observed universe. That cosmology does not yet exist.

5 The Dark Matter Slot

5.1 Two Fold Sectors

The fold framework currently treats all nodes identically: same Φ , same $\bar{\tau}$, same κ . A natural extension is two populations:

- **Visible sector:** nodes with threshold κ_v , coupling to the electromagnetic fold cycle (Papers 1–2).
- **Dark sector:** nodes with threshold $\kappa_d \gg \kappa_v$, no electromagnetic coupling.

High- κ nodes accumulate $\bar{\tau}$ longer before resetting. They form deeper potential wells. They cluster more readily. They do not participate in the color-neutrality attractor (Paper 2) because they never reach the 3-state cycle threshold.

5.2 Why It Arrives with Shell Closure

Dark sector nodes are present in the fold field from the beginning — they are simply fold nodes with high κ . But they become gravitationally significant only after shell closure, when the inverse-square law (Paper 3) begins operating. Before 3D space exists, there is no distance, no $1/r^2$, no gravitational clustering.

After shell closure, dark sector nodes cluster first (deeper wells, longer accumulation, stronger local g) and then visible matter falls into the resulting structure. This is the standard dark matter picture — but derived from fold dynamics rather than postulated.

5.3 Honest Status

This is a candidate identification. The two-sector fold field has not been simulated. The coupling between sectors (through the shared gravitational field) has not been derived. This is noted here as the natural dark matter slot in the framework, to be developed in future work.

6 Quaoar as a Shell Boundary Test

The Quaoar ring anomaly (Paper 3, Open Problem discussion) is now interpretable as a shell boundary effect. The ring sits at the radius where Quaoar’s local fold field and the solar system’s external field are in balance. Inside this radius: material is in Quaoar’s shell, experiences Quaoar’s B , accretes normally. Outside this radius: material is in the solar system shell, subject to tidal disruption.

The ring persists at the boundary because:

1. $g_{\text{local}} \approx g_{\text{external}}$ — neither dominates, so neither accretes nor disrupts completely.
2. $B_{\text{Quaoar}} \approx B_{\text{solar}}$ at the boundary — the B -differential is near zero, reducing the effective tidal pressure.

This is a testable prediction: ring systems should preferentially form near the Hill sphere boundary, not at the classical Roche limit. The classical Roche limit assumes a pure $1/r^2$ field without the B term; the fold Roche limit includes the B -differential and predicts a shifted stability boundary.

6.1 Connection to the Fold Scaling Kernel

A companion paper [9] derives the fold ring stability condition from the lock condition $T_{\text{fold}}(r) = T_{\text{orb}}(r)$. This yields the renormalization relation:

$$\left(\frac{r}{l_P}\right)^n = \frac{\kappa_{\text{eff}}}{\kappa_{\text{sim}}}, \quad (14)$$

where $\kappa_{\text{eff}} = 2\pi v_{\text{orb}}/c$ and $\kappa_{\text{sim}} = 2.0$ is the Planck-scale simulation threshold. This relation describes precisely how the fold threshold κ is renormalized as you scale from the Planck length to macroscopic ring radii.

This is the Module 3 shell hierarchy in quantitative form: the effective fold threshold at any scale is set by the ratio $(r/l_P)^n$, where n is measured from the ring system’s body type (giant planet: $n \approx -0.093$; TNO: $n \approx -0.135$; Centaur: $n \approx -0.160$). Seven ring systems across four body types follow this three-population structure, providing the first observational constraint on how fold dynamics scale from Planck to planetary scales.

The scale-dependent B field of Module 3 and the scale-dependent κ_{eff} of the ring kernel are two aspects of the same underlying physics: fold dynamics that weaken with scale in a structured, measurable way.

7 Open Problems

Open Problem 1 (Fold Cosmology). The magnitude of B_{boundary} and its connection to the observed dark energy density requires a fold cosmological model — expansion dynamics, scale factor evolution, and the T3 conservation law applied at cosmological scales. This is the most important open problem in the present work.

Open Problem 2 (Scale Invariance Proof). Section 2.3 provides a nondimensionalization argument showing that the shell boundary condition is genuinely scale-free (not merely lacking explicit scale). What remains unproven is whether the full dynamical solution — including initial conditions and boundary conditions at each scale — preserves this invariance. A simulation confirming that Module 2.3 shell formation dynamics produce the same qualitative behavior (same $C = 1.014$ plateau, same transport suppression) at widely separated physical scales (e.g. $r \sim l_P$ and $r \sim 10^{35} l_P$) would close this problem.

Open Problem 3 (Dark Matter Sector). The two-sector fold field (visible κ_v + dark κ_d) needs simulation and analytical development. Key questions: what determines κ_d/κ_v ? How does the dark sector cluster differently? Does the ratio of dark to visible matter density fall out of fold dynamics, or is it a free parameter?

Open Problem 4 (Quaoar Simulation). The shell boundary prediction for ring systems needs a simulation: run fold diffusion around a point mass within an external field, measure where the B-differential drops to near zero, and check whether ring stability occurs at that radius rather than the classical Roche limit.

8 Summary

The shell-formation dynamics of Module 1 extend naturally to all scales. The shell boundary is the surface where local and external fold fields balance (Definition 2.1). The background term B screens completely inside each shell, becoming the local reference. The cosmological shell — closed during the 2D→3D transition — has no external reference, so its B_{boundary} is anchored at shell-closure conditions.

As structure forms, B_{interior} depletes. The growing differential $\Delta B = B_{\text{boundary}} - B_{\text{interior}}$ acts as an outward pressure without requiring $B < 0$ — only that the interior is lower than the boundary, as in a closed pressurized system. This is the fold dark energy candidate.

The dark sector arrives late because it cannot exist before shell closure. The coincidence problem is resolved because dark energy is sourced by the same structure formation that produces visible matter clustering.

The fold shell hierarchy:

Shell boundary: $g_{\text{local}}(r^*) = g_{\text{ext}}(r^*)$ (bias to the larger)

B -screening: $B_\ell = (M_{\text{total},\ell} - M_{\text{structure},\ell})/V_\ell$ (T3 within shell)

$\Delta B = B_{\text{boundary}} - B_{\text{interior}} \geq 0$ (grows as structure forms)

$\Delta B \rightarrow B_{\text{boundary}}$ as $a \rightarrow \infty$ ($w \rightarrow -1$ at late times)

Dark energy is the B-differential of a closed fold system.

It cannot appear before the shell closes.

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Code and data available at: <https://github.com/PrimeFoldTheory/foundations>

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